Problem FoL.1 ... underground

3 point

Matěj, with mass $m = 70\,\mathrm{kg}$, entered an underground train car and spotted his friend Jáchym at the other side of the train car, at distance $s = 20\,\mathrm{m}$. Matěj started walking towards Jáchym in the direction of motion of the train. The train was moving with constant acceleration $a = 1\,\mathrm{m\cdot s^{-2}}$. How much work must Matěj do to reach Jáchym?

Matěj loves travelling by underground

Problem FoL.2 ... passing trains

3 point

Danka was watching a train with length $l_1 = 120\,\mathrm{m}$, *which was passing her with velocity* $v_1 =$ $= 60\,\mathrm{km{\cdot}h^{-1}}$. *She also saw a train of unknown length* l_2 *approaching from the opposite directio with velocity* $v_2 = 80\,\mathrm{km{\cdot}h^{-1}}$. *In order to find* l_2, *Danka measured the time the trains spen passing each other, i.e. the interval between the times when the fronts of the trains met an when the backs of the trains met. She measured* $t = 9\,\mathrm{s}$. *How long was the second train?*

Danka was waiting for a train

Problem FoL.3 ... let's play a game

3 point

You are playing a game which consists of separate rounds. When you lose a round, you los
point. When you win a round, you gain a point, and if you have also won the previous tw
ounds, you gain another point as a bonus. If the chance of winning a round is 50 % (draw i
ot possible), what is the long-term average (expected value) of the number of points gaine
er round?

Jáchym plays only ranked matches in Hearthstone

Problem FoL.4 ... Keep pulling or I'll whip you! 3 point

When building the pyramids, the transport of giant stone blocks was done by exploitation o he good old slave labor. The blocks are transported on an inclined plane with slope $\alpha = 15^\circ$ Whipmaster Jáchym found out that to pull a block up the inclined plane, the full force of 1 laves is needed. However, when transporting the same block down, he only needs five slave ulling with full power. Determine the friction coefficient between the block and the incline lane. Each slave is equally strong and pulls a rope parallel to the inclined plane with constan orce.

Matěj was interested in pulling of blocks

Problem FoL.5 ... Theophist temperature

3 point

n hell, temperature is measured on the Theophist scale. The conversion formula betwee
Theophist and Celsius scales is linear. The temperature 100 °T *(degrees Theophist) is th*
oiling temperature of sulfur 445 °C*. Once it was so cold in hell that paraffin wax solidifie*
At that time, the hell's meteorological station measured −62 °T*, which corresponds to* 40 °C
*Determine the boiling temperature of mercury (*357 °C*) in hell's units* °T*.*

You are going to get into hot mercury about tha

Problem FoL.6 ... stairs

3 point

Danka steps on the top stair of an escalator. When she is standing still, the escalator carrie
her to the bottom in $t_1 = 2.0\,\mathrm{min}$. *If the escalator was not moving and Danka walked down it*
t would take her $t_2 = 2.5\,\mathrm{min}$ *to get to the bottom. How long will it take her to get from th*
op to the bottom if she starts walking down the moving escalator in the middle of the path
Danka walks with constant velocity. *Danka stopped to think in an undergroun*

Problem FoL.7 ... hockey-football problem

4 point

There is a radioactive substance with N_0 particles. Let's denote the number of particles tha decay during the second $1/3$ of the first half-life by ΔN. Find the ratio $\Delta N/N_0$.

Waiting for the third third of Half-Life

Problem FoL.8 ... resistive block

3 point

We have $l = 11.4\,\mathrm{m}$ of wire with circular cross-section with diameter $d = 0.61\,\mathrm{mm}$. The resis
ance of the wire is $R = 19.6\,\Omega$. What would be the resistance R_a of a cube formed by smeltin
his wire, if it was measured under direct current, between two perfectly conductive plate
ouching two opposite sides of the cube? *Karel was thinking up simple physics problems*

Problem FoL.9 ... queen's justice

3 point

Poor serfs are bringing eggs to the queen and receive gold coins in exchange. Each egg is worth ten gold coins. However, if a serf brings eggs worth more than 6,700 *gold coins in total, the queen will pay* 1,600 *gold coins less for them to make sure landowners won't get as rich as Her Majesty. If they bring eggs worth from* 5,000 *to* 6,700 *gold coins (inclusive), the pay is reduced only by "only"* 1,000 *gold coins.*

A farmer is breeding hens that lay from 0 to 1,000 *eggs per week (uniformly randomly distributed), which he brings to the queen each Sunday. What is the average (expected) amount of gold coins he earns per week? Matěj and Jáchym are obedient servants of Her Majesty*

Problem FoL.10 ... scuba diver

4 point

A scuba diver is $h = 10\,\mathrm{m}$ under the water surface. How large is the area of the surface o
which he can see the sky? The refractive index of water is 1.33 *and the refractive index of ai*
s 1.00.

Šimon is reminiscing about summe

Problem FoL.11 ... fifty

3 point

Vašek calibrated his tachometer so that it would show the exact velocity of his car based o
the rotation speed of the wheels. However, his tyres are worn and their radius is 35.7 cm*, so h*
eplaces them by new ones with 4 mm *larger radius, but forgets to recalibrate the tachometer*
How much (in $\mathrm{km\cdot h^{-1}}$*) does his real velocity differ from the velocity displayed on the tachome*
er, which is $50\,\mathrm{km\cdot h^{-1}}$*?*

Vašek follows traffic laws

Problem FoL.12 ... Watch out for the turn!

3 point

A car with mass $m = 1.50\,\mathrm{t}$ is taking a turn with radius $R = 30.0\,\mathrm{m}$. The static friction coefficient between the tyres and road surface is $f = 0.55$. With what maximum velocity may the car take the turn without skidding?

Danka took a turn too fast

Problem FoL.13 ... Planck linear density

5 point

Planck units are physical quantities formed as suitable combinations of speed of light c = $= 3.00 \cdot 10^8\,\mathrm{m{\cdot}s^{-1}}$, *reduced Planck constant* $\hbar = 1.05 \cdot 10^{-34}\,\mathrm{kg{\cdot}m^2{\cdot}s^{-1}}$ *and the gravitationa constant* $G = 6.67 \cdot 10^{-11}\,\mathrm{kg^{-1}{\cdot}m^3{\cdot}s^{-2}}$. *That means we may write* $A = c^\alpha \cdot \hbar^\beta \cdot G^\gamma$, *obtainin all sorts of physical quantities. Consider such an expression for linear density, which is usuall denoted by* λ *and has units* $\mathrm{kg{\cdot}m^{-1}}$. *Compute the sum of exponents* $\alpha + \beta + \gamma$.

Karel likes to think about Planck's units

Problem FoL.14 ... drop velocity

3 point

Compute the velocity of a raindrop (a sphere with radius $3\,\mathrm{mm}$*) immediately before hittin the ground. Assume the flow of air around a raindrop is turbulent. The density of air is* $\varrho = 1.25\,\mathrm{kg{\cdot}m^{-3}}$ *and the drag coefficient of a spherical raindrop is* $C = 0.5$*.*

Katarína likes walking in the rain

Problem FoL.15 ... giga-cave

5 point

Consider a perfectly spherical planet with radius R. The planet is homogeneous, except fo a spherical cave with radius R/8 touching the surface of the planet. What is the ratio o gravitational acceleration close to the surface at the point where the cave touches the surfac to gravitational acceleration at the opposite point? *Karel likes spherical planets*

Problem FoL.16 ... compound bow or semiautomatic pistol? 5 point

We fire an arrow from a compound bow PSE X-Force Omen and a bullet from a semi-automati pistol Colt 1903 .32 directly against each other. If these two projectiles collided in a perfectl nelastic collision and merged together, what would be the velocity (including the direction) o he resulting projectile? The arrow has mass $m_1 = 350\,\mathrm{grain}$ *(as you surely know,* $1\,\mathrm{grain} = 6.479 \cdot 10^{-5}\,\mathrm{kg}$*) and velocity* $v_1 = 366\,\mathrm{ft{\cdot}s^{-1}}$ *(feet per second,* $1\,\mathrm{ft{\cdot}s^{-1}} = 0.3048\,\mathrm{m{\cdot}s^{-1}}$*). Th bullet,* $7.65\,\mathrm{mm}$ *Browning Short, has mass* $m_2 = 65\,\mathrm{grain}$ *and velocity* $v_2 = 925\,\mathrm{ft{\cdot}s^{-1}}$*. Assum non-rotating point projectiles. The answer should be positive if the resulting projectile flies i he direction of the arrow or negative if it flies in the direction of the bullet.*

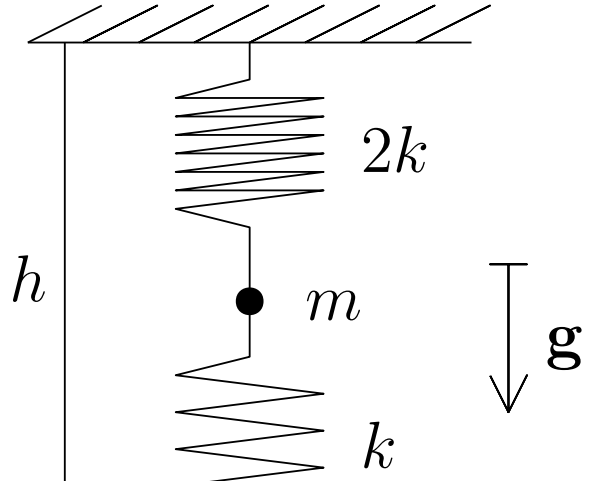
Karel was thinking about ranged weapons

Problem FoL.17 ... twice as stiff spring

3 point

There are two springs with negligible masses and one point mass in the configuration depicted in the figure. The stiffness of the lower spring is $k = 5.0\,\mathrm{kg{\cdot}s^{-2}}$ *and the stiffness of the upper spring is* $2k$*. The rest length of each spring is* $l_0 = h/4 = 10\,\mathrm{cm}$ *and they are both stretched to lengths* $2l_0$*. What should be the mass of the point mass if we want the whole configuration to be in equilibrium? The acceleration due to gravity is* $g = 9.81\,\mathrm{m{\cdot}s^{-2}}$*.*

Karel varied Paťo's problem from Slovak Physics Olympiad.



Problem FoL.18 ... a warm-up hurricane

4 point

The main driving force of a cyclone is the ocean, or more specifically, the heat lost by wate evaporating from the surface of the ocean. Steam particles are quickly lifted up to the uppe part of the cyclone, where water condensates due to lower temperature, pressure and presence o condensation nuclei. In the last part of the cycle, it falls down to the ocean surface as rain (thi process is also very fast). Therefore, the cyclone works between two temperatures – the ocea temperature $T_1 = 280\,\mathrm{K}$ *and the cloud temperature* $T_2 = -26\,^\circ\mathrm{C}$*. It turns out that this proces can be described as a thermodynamic cycle with the ideal gas. Using the given information compute the efficiency of a cyclone as a thermodynamic heat engine with water as the workin substance.*

Vítek watched news

Problem FoL.19 ... rain

3 point

t started raining and the Fykos bird noticed that the density of raindrops (hits per unit area) o
he western side of the faculty building is twice as large as on the southern side. Determine th
direction of wind – specifically, the angle between this direction and the northward direction
n degrees. Assume the sides of the building are exactly facing the four cardinal directions.

Matěj likes to sit inside college during the rain

Problem FoL.20 ... throwing balls

4 point

Mišo and Dano are standing next to each other. At the same time, each of them throws a tennis ball with the same velocity $v = 15\,\mathrm{m\cdot s^{-1}}$. *Dano throws his ball under an elevation angle* $\alpha_1 = 45^\circ$ *and Mišo throws his under an elevation angle* $\alpha_2 = 35^\circ$. *How much farther will the winning ball go? Assume that the balls fly next to each other in parallel planes and start from zero height. Neglect air resistance.*

Danka remembered physical education lessons

Problem FoL.21 ... shadow

4 point

What is the smallest (absolute) latitude such that when you are standing on a horizontal surfac
t this latitude, your shadow (created by the Sun) will never be shorter than you?

Matěj is hiding in the shadows of the others

Problem FoL.22 ... elevator acceleration

4 point

An elevator should travel from the ground level of a building to the top at height 60 m. *During the acceleration phase, its acceleration must not exceed* $4\,\mathrm{m\cdot s^{-2}}$; *when it decelerates, it must not exceed* $6\,\mathrm{m\cdot s^{-2}}$ *(in absolute value). What is the minimum time the elevator needs to reach the top of the building? Are there any limits to the optimization of the elevator usage*

Problem FoL.23 ... Kevin the Cube

5 point

A large homogeneous cube with side length $a = 15\,\mathrm{m}$ decided to roll around one of its edges on a horizontal surface in such a way that the energy used for this would be the minimum possible. Unfortunately, at distance a from this egde, there is a tomato that is soon to be squashed. With what velocity will an edge of the cube hit the tomato? The acceleration due to gravity is $g = 9.81\,\mathrm{m\cdot s^{-2}}$.

Matěj is fascinated by rolling balls

Problem FoL.24 ... chained

5 point

A chain with mass 6.0 kg *and length* 11 m *is hanging down from one end at height* 15 m *abov the ground. Find the maximum kinetic energy of the chain after it is released. Consider tha the chain hits the ground perfectly inelastically.*

Josef Jírů wanted to enrich our participants with a problem with a chain

Problem FoL.25 ... cork problem

4 point

Vítek was investigating some fluid properties using a water surface with surface tension $\sigma_1 =$ $= 70\,\mathrm{mN{\cdot}m^{-1}}$. *For this purpose, he made a thin cork board with mass* $m = 100\,\mathrm{g}$ *and sid lengths* $b = 5\,\mathrm{cm}$ *and* $c = 7\,\mathrm{cm}$, *which was – to his surprise – floating on the surface. H decided to get the board moving, so he added a bit of detergent to one of its shorter sides. Th surface tension of the resulting soap (detergent) solution is* $\sigma_2 = 40\,\mathrm{mN{\cdot}m^{-1}}$. *Help Vítek fin the acceleration of the cork board. For simplicity, assume that water resistance is negligible an that the contact angles are always* $90°$. *Vítek likes to accelerate cork with some detergen*

Problem FoL.26 ... olive one

6 point

An ideal, seedless olive may be modelled by a hollow sphere with inner and outer radi $r_1 = 3.00\,\mathrm{mm}$ *and* $r_2 = 10.0\,\mathrm{mm}$, *respectively. If the density of an ideal olive is* $\varrho_\mathrm{o} = 816\,\mathrm{kg{\cdot}m^{-3}}$ *how deep will it submerge in brine with density* $\varrho_\mathrm{b} = 1{,}211\,\mathrm{kg{\cdot}m^{-3}}$*? Assume that an ideal oliv is perfectly tasty and that brine fills the hole created by the missing seed.*

Jáchym is attracted by olives

Problem FoL.27 ... Paťo is doing sports

5 point

Legend says Paťo finished a twelve-minute run in six minutes. Along the track, there are mile stones at equal distances $\Delta l = 5.000\,000\,000\,\mathrm{km}$*; the first milestone is* Δl *from the starting point. Find the number of milestones Paťo passed if his watch showed that he ran for six minutes when he finished the run. The speed of light is* $299{,}792{,}458\,\mathrm{m\cdot s^{-1}}$*.*

Every evening, Lukáš looks under his bed to make sure Paťo isn't there

Problem FoL.28 ... energetic pendulum

5 point

A mathematical pendulum is a small ball with mass $m = 5\,\mathrm{g}$ *hanging from a (massless) rope wit*
ength $l = 2\,\mathrm{m}$. *We displace the ball from its equilibrium position by an angle* $\alpha = 4^\circ 10'$ *and le*
t oscillate. After completing one swing (half-oscillation), the ball retains $99\,\%$ *of the mechanica*
nergy it had before that swing. After how many full swings (half-oscillations) will the velocit
f the ball when passing through the equilibrium position decrease under $v_h = 0.2\,\mathrm{m{\cdot}s^{-1}}$
Assume the pendulum loses its energy symmetrically with respect to the equilibrium position

Daniela closely watched the "renovation" of student cafeteria in Troja

Problem FoL.29 ... Einstein with Cherenkov

6 point

A photon collides with a gold surface submerged in a medium with refractive index $n = 1.804,1$ *and ejects an electron. This electron is so fast that in this medium, it is moving faster tha the speed of light in this medium. What is the maximum possible wavelength of the pho on? The work function of gold is* $W_0 = 5.45000\,\mathrm{eV}$. *The speed of light in vacuum is* $c =$ $= 2.997,925 \cdot 10^8\,\mathrm{m{\cdot}s^{-1}}$, *Planck constant* $h = 6.626,07 \cdot 10^{-34}\,\mathrm{J{\cdot}s}$, *electron rest mass* $m_0 =$ $= 9.109,38 \cdot 10^{-31}\,\mathrm{kg}$ *and elementary charge* $e = 1.60218 \cdot 10^{-19}\,\mathrm{C}$.

Karel was looking into a nuclear reacto

Problem FoL.30 ... two-state problem

5 point

The Fykos bird started learning statistical physics, but he cannot solve the following introduc ory problem. Consider N particles in a closed box. Let's split them into two groups – particle n the right half of the box are in state 0 and those in the left half are in state 1. We do no are about which individual particles have which state, only that N_0 have state 0 and N_1 hav tate 1 ($N_1 + N_0 = N$). Microstates of the system are described by a number $m = N_1 - N_0$ which describes the distribution of particles between the two states. Let's denote the numbe of ways to split the particles into the two states by $t(m)$ (e.g. for $N = m = 10$, $t = 1$). Not hat the exact analytical formula may be simplified to

$$t(m) = \sqrt{\frac{2}{\pi N}} 2^N \exp\left(-\frac{m^2}{2N}\right)$$

using Stirling approximation. Evaluate both formulas (exact and approximate) for $N = 20$ an $n = 8$ and find the relative error (in percentage) of the approximate value with respect t he exact value, i.e. by how many percent the approximate value differs from the exact one

Vítek is in another state

Problem FoL.31 ... spacey

5 point

n a lab on Earth, we measure the radiation spectrum of a star that is moving away from Eart with velocity $v = 1.70 \cdot 10^7\,\mathrm{m{\cdot}s^{-1}}$*. The maximum intensity in this spectrum is at wavelengt* $\lambda_0 = 700\,\mathrm{nm}$*. Determine the surface temperature of the star. Assume that the star is an idea black body. The speed of light is* $c = 3.00 \cdot 10^8\,\mathrm{m{\cdot}s^{-1}}$*.*

Vítek was reading Fundamental Astronomy

Problem FoL.32 ... one-third train

3 point

A train was moving with velocity $96\,\mathrm{km\cdot h^{-1}}$*. It engaged its brakes for the first time at time* 99
before coming to a halt. What was the distance at which it engaged the brakes? During th
first 1/3 *of this distance, the train was decelerating with constant acceleration and decreased*
its velocity to 1/2*. During the second* 1/3 *of the same distance, it was moving with constan*
velocity, and during the last 1/3*, it was again decelerating with constant acceleration until i*
stopped. *Josef Jírů wanted to enrich our participants with a problem with trains*

Problem FoL.33 ... marble on a spiral

5 point

There is a spiral defined by parametric equations $x = \sin\varphi$, $y = -\cos\varphi$, $z = \varphi/2\pi$, *placed in a homogeneous gravitational field* $g = 9.81\,\mathrm{m\cdot s^{-2}}$ *in the direction of the negative* z*-axis. We place a marble on the spiral. The marble does not rotate and moves on the spiral without friction. What will the negative* z*-component of its acceleration be?*

Karel likes to play with marbles

Problem FoL.34 ... olive two

6 point

An ideal olive is a perfect sphere containing a spherical seed with radius $r_1 = 3\,\mathrm{mm}$. *The densit of an ideal olive is equal to the density of brine* $\varrho_\mathrm{n} = 1{,}211\,\mathrm{kg{\cdot}m^{-3}}$ *so that it would freely floa in a can. Consider an open cylindrical can of olives in brine with base radius* $R = 31\,\mathrm{mm}$. *If w remove the seed from an olive, the rest of the olive floats up to the surface of the brine and thi surface decreases by* $\Delta h = 4 \cdot 10^{-5}\,\mathrm{m}$ *in the process. What is the density of the seed? Assum that the brine fills the hole created by removing the seed, too.*

Jáchym cannot help himself and still thinks just about olives

Problem FoL.35 ... Dan's eggs

7 point

Dan wants to throw a raw egg to Danka in such a way that she can catch it at minimal velocit
(in order to minimise the probability that it breaks when caught). He's throwing from a balcon
at height h, Danka is at height 0. Due to bad terrain, Danka can only stand at a distance greate
than $3h$ from the base of the balcony. Find the elevation angle under which Dan should throw
the egg. Do not consider air resistance.

Girls like to grab eggs

Problem FoL.36 ... a motivational capacitor

6 point

Vítek has a favourite parallel-plate capacitor. Its plates are separated by distance $d = 2.0\,\mathrm{mm}$ *and the area of each plate is* $S = 20.0\,\mathrm{cm}^2$. *He placed a charge of* Q *on one plate and a charge of* $-Q$ *on the other plate, where* $Q = 0.1\,\mathrm{nC}$. *Then he decided to fill the space between the plates with a dielectric material. As a result, the electric intensity between the plates is no longer constant, but given by the formula*

$$E(x) = -\frac{Q\mathrm{e}^{x/d}}{\varepsilon_0 \kappa_0 S}\,,$$

where $\kappa_0 = 1.5$ *and* x *is the distance from the negatively charged plate. Find the capacitance of this capacitor.*

Vítek broke a capacitor at his practical class

Problem FoL.37 ... a calm place

5 point

In a vast, empty country, there are two very long, straight, parallel roads $s = 490.0\,\mathrm{m}$ *apart Matěj found himself between these two roads. However, he cannot cross the first road, becaus there are* $n_1 = 0.90$ *cars per second flowing through it. The second road has even more traffic cars flow through it at a rate of* $n_2 = 1.60$ *cars per second. The velocity of cars is the sam on both roads. He had no choice but to wait for death, since there was no crossing in sight Therefore, he found a place where the total noise from both roads was at a minimum, sat down and waited... How far from the first road is he sitting? There are no sound barriers, soun travels through the air isotropically and without losses and each car is equally noisy.*

Matěj got this idea when he wanted to kill himself after not acing an exam

Problem FoL.38 ... finally a complex infinite circuit

5 point

Consider the infinite RLC circuit in figure. The inductance of each coil is $L = 3\,\mathrm{mH}$*, the capacit of each capacitor is* $C = 4{,}700\,\mathrm{nF}$ *and the resistances are* $R_1 = 1.5\,\mathrm{k\Omega}$ *and* $R_2 = 0.3\,\mathrm{k\Omega}$*. A which frequency of alternating current is the impedance of the circuit purely real? Neglect phas shift caused by finite propagation speed of electromagnetic field.*

Jáchym likes to be complex

Problem FoL.39 ... vertical shelf

4 point

There is a copper cuboid with length of $l_0 = 1{,}970.0\,\mathrm{mm}$ *and mass of* $m = 79.00\,\mathrm{kg}$*. If th emperature of copper decreases by* $\Delta T = 12\,\mathrm{K}$*, the cuboid perfectly fits between two rigi blocks (which do not expand or contract with changes in temperature). We wait for the coppe o heat up to its original temperature. What mass may we place on the cuboid before it fall down? The static friction coefficient between copper and the blocks is* $f = 0.15$*. Assume tha he cuboid does not deform when the mass is placed on it.*

Jáchym thought up a new way to store things

Problem FoL.40 ... olive three

5 point

Jáchym is eating olives with eating speed proportional to the square of the number of uneate olives. A day after he opened a can, there were 32 of them. A day later, there were just 17 left How many olives were in the can initially? Assume that the number of olives is a continuou variable for Jáchym.

Jáchym dreams only about olives

Problem FoL.41 ... elastic loop

6 point

A circular loop with radius $R_0 = 0.45\,\mathrm{m}$ is placed in a homogeneous static magnetic field wit magnitude $B = 0.83\,\mathrm{T}$ in such a way that its axis is parallel with the direction of the magneti field and the loop is compressed by the manetic field. There is an electric current $I = 16\,A$ lowing through the loop. The loop is made from a brass wire with circular cross-section (wit adius $r = 0.254\,\mathrm{mm}$) and Young's modulus $E = 98\,\mathrm{GPa}$. How does the length of the loo educe (in mm*)?*

Vašek wanted to know what happens to a magnet in a magnetic field



Problem FoL.42 ... astronaut training

5 point

There is a swing carousel standing on the Earth's surface (in gravity g). With what frequenc should it turn so that the visitors riding it would experience net acceleration $2g$? The carouse may be considered a circle with radius $R = 3.5\,\mathrm{m}$ placed $H = 4\,\mathrm{m}$ above the ground with seat ttached to the perimeter of this circle by chains. Let's assume that a person sitting in a seat i point mass at distance $l = 3\,\mathrm{m}$ from the point where the seat is attached to the circle, chain nd seats are massless, chains have constant lengths and able to rotate freely around the point where they're attached to the circle. Neglect resistive forces.

Karel was watching the Kosmo series

Problem FoL.43 ... exotic field

6 point

Vítek found a very exotic field in an old textbook. Consider a point-like mass $m = 1.00\,\mathrm{k}$ *at rest in equilibrium in the potential* $V(x) = -Ax^{10}\exp(Bx)$*, where* $A = 2.00\,\mathrm{kg{\cdot}m^{-8}{\cdot}s^{-}}$ *and* $B = 0.01\,\mathrm{m}^{-1}$*. The mass is then slightly displaced from its equilibrium position. Fin the period of small oscillations of the point-like mass around the equilibrium position in thi potential field.*

Vítek is imagining how Matěj is conducting experiments

Problem FoL.44 ... identifying a rod

8 point

Vítek had a cylindrical wire with radius $R = 1.20\,\mathrm{cm}$ *made from an unknown material connecte o a DC source. Luckily, he knew the energy density of magnetic field* $w_{\mathrm{mag}} = 0.50\,\mathrm{J{\cdot}m^{-3}}$ *an he energy density of electric field* $w_{\mathrm{el}} = 2.50 \cdot 10^{-17}\,\mathrm{J{\cdot}m^{-3}}$ *immediately above the surface of th onductor. Both fields are parallel with the surface of the conductor. Moreover, using a magne Vítek found out that the rod is non-magnetic. The conductor is placed in air, the Hall effect i gnored. Find the conductivity of the cylinder's material.*

Vítek wanted to make a problem that just needs standard formulas

Problem FoL.45 ... doublets

8 point

We used a sodium lamp to symmetrically illuminate two $b = 1.0\,\mu\text{m}$ *wide slits* $a = 0.10\,\text{mm}$ *apart. The light passing through the slits is observed on a screen at distance* $d = 1.0\,\text{m}$. *Determine the ratio of intensity of incident light in the* 1st *minimum to the intensity in the* 0th *maximum.*

Assume that a sodium lamp emits light only on two wavelengths $\lambda_1 = 589.0\,\text{nm}$ *and* $\lambda_2 = 589.6\,\text{nm}$ *with intensity ratio 2:1.*

Xellos likes to enlighten people

Problem FoL.46 ... stellar sail

8 point

At a distance $d = 100\,\mathrm{AU}$ *from the Sun, there is a mirror with area* $S = 100\,\mathrm{km}^2$ *moving awa from the Sun with velocity* $v = 0.4c$*. The mirror is quite strange – it completely reflects ligh in the range* $\langle 400\,\mathrm{nm}, 500\,\mathrm{nm}\rangle$ *and is completely transparent for all other wavelengths. Fin the force due to solar radiation acting on the mirror, according to an observer at the mirror Assume that the Sun is an ideal black body with surface temperature* $T = 6000\,\mathrm{K}$ *and radiu* $R = 7 \cdot 10^5\,\mathrm{km}$*.*

Xellos would like to have a Solar Sailer

Problem FoL.47 ... throw over a cylinder

6 point

There is a huge cylinder with base radius $R = 10\,\mathrm{m}$ lying on its side. You want to throw a ston from ground level so that the stone flies over the cylinder. What is the minimum required initia velocity of the stone? Jáchym guessed the original task from a figure in one spanish book

Problem FoL.48 ... evil spring

9 point

Consider a homogeneous spring with rest length $l = 11.3\,\mathrm{cm}$, *stiffness* $k = 125\,\mathrm{kg{\cdot}s^{-2}}$ *and linea density* $\lambda = 9.7\,\mathrm{kg{\cdot}m^{-1}}$ *placed on a horizontal pad. We fix one of its endpoints in place in suc way that the spring can still freely rotate around this endpoint. Then, we remove the pad Calculate the difference between the total potential energy of the spring in its new equilibriu position and in the initial position.* *Jáchym couldn't figure out the principle of a pen*

Problem M.1 ... Dyson

3 point

A Dyson sphere is a (so far) hypothetical construction around a star which completely surround the star in order to capture all the energy it radiates. Imagine that someone takes our Earth and creates a Dyson sphere from its matter, with our Sun at the centre of the sphere. The outer radius of this sphere is the same as the average Earth-Sun distance $d_S = 149\,600\,000\,\mathrm{km}$. *Assume that Earth is a sphere with radius* $R_Z = 6\,378\,\mathrm{km}$ *and that its matter is homogeneous and incompressible – therefore, the density of the body of the Dyson sphere is the same as the density of Earth. What is the thickness of this Dyson sphere?*

Karel likes problems with a Dyson sphere

Problem M.2 ... Dyson Reloaded

3 point

Let's consider the Dyson sphere from the previous problem. (Imagine that someone takes ou Earth and creates a Dyson sphere from its matter, with our Sun at the centre of the sphere. Th outer radius of this sphere is the same as the average Earth-Sun distance $d_S = 149\,600\,000\,\mathrm{km}$ *Assume that Earth is a sphere with radius* $R_Z = 6\,378\,\mathrm{km}$ *and that its matter is homogeneou and incompressible – therefore, the density of the body of the Dyson sphere is the same a the density of Earth.) Calculate the gravitational acceleration that could be felt by astronaut standing on its outer surface. The mass of Earth is* $M_E = 5.97 \cdot 10^{24}\,\mathrm{kg}$ *and the mass of the Su is* $M_S = 1.99 \cdot 10^{30}\,\mathrm{kg}$.

Karel really likes problems with a Dyson sphere

Problem M.3 ... Dyson Reloaded Reloaded

3 point

For now, let's keep discussing the same Dyson sphere. (Imagine that someone takes our Earth and creates a Dyson sphere from its matter, with our Sun at the centre of the sphere. The outer radius of this sphere is the same as the average Earth-Sun distance $d_S = 149\,600\,000\,\mathrm{km}$*. Assume that Earth is a sphere with radius* $R_Z = 6\,378\,\mathrm{km}$ *and that its matter is homogeneous and incompressible – therefore, the density of the body of the Dyson sphere is the same as the density of Earth.) We already know that it would be hard to walk on its outer surface, but could we walk on its inner surface? How many days would a month (i.e.* 1/12*-th of one orbit around the Sun) take, if the sphere rotated with such an angular velocity that the acceleration*

due to gravity on its equator would be the same as on the equator of the (regular) Earth?

Karel really likes problems with a Dyson sphere

Problem M.4 ... Dyson Reloaded[3]

3 point

Let's trouble our minds with the Dyson sphere one last time. I promise it's the last time Imagine that someone takes our Earth and creates a Dyson sphere from its matter. The outer adius of this sphere is the same as the average Earth-Sun distance $d_S = 149\,600\,000\,\mathrm{km}$ *Assume that Earth is a sphere with radius* $R_Z = 6\,378\,\mathrm{km}$ *and that its matter is homogeneous nd incompressible – therefore, the density of the body of the Dyson sphere is the same as the lensity of Earth.) In this case, let's assume that the engineers made a mistake somewhere and lid not build the sphere around the Sun. The sphere stays the same (it has the same radius nd it is made of the Earth's matter), it does not rotate and there are no stars or planets on he inside or anywhere nearby. What is the minimum amount of pressure (critical stress) that naterial of Earth must be able to withstand so that the sphere does not collapse under its own veight?*

Matěj became infected and now he also likes Dyson spheres

Problem E.1 ... oh, these batteries

3 point

Imagine that you have a fully discharged car battery ($U = 12\,\mathrm{V}$, capacity $60\,\mathrm{Ah}$) and a chargin
device that can supply the battery with a steady current $I = 3.0\,\mathrm{A}$. If you can charge the batter
for $t = 75\,\mathrm{min}$, what will its final charge level be (in percent of total capacity)?

Karel. Don't even ask..

Problem E.2 ... inner resistance

3 point

f we connect two identical resistors in series to a non-ideal DC voltage source, the efficiency o
he source is 0.87*. What is the efficiency of the source if we connect the two resistors to it i*
arallel?

We struggled with a revolution

Problem E.3 ... solar panel incident

3 point

A horizontal photovoltaic panel generates power 300 W *when sunlight is incident on it perpen*
dicularly. What power would it generate at 10:00, on a day when the Sun rises at 6:00 and set
at 18:00? At noon, the angle of incidence of sunlight on the panel is 20° *from the vertical.*

Eating green wasabi, remembering green energy, improperly writing gerunds

Problem E.4 ... transforming

3 point

Consider a transformer that has a primary coil with $N_1 = 100$ turns and a secondary coil wit $N_2 = 300$ turns. A light bulb with resistance $R = 25.0\,\Omega$ is connected to the secondary coi We want the light bulb to be supplied with power $P = 200\,\mathrm{W}$. What should be the effectiv voltage connected to the primary coil of the transformer?

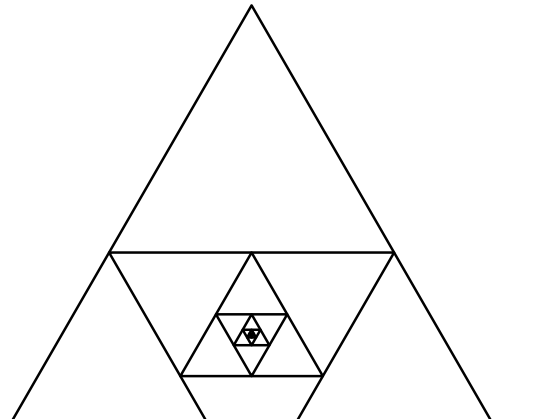
Danka was reminiscing about high school problems

Problem X.1 ... purely fractal

3 point

Matěj likes fractals, so he made one out of infinitely thin wire with linear density $\lambda = 100\,\mathrm{g\cdot m^{-1}}$. He started by making an equilateral triangle with side length $l = 10\,\mathrm{cm}$, added the midsegments of this triangle, then added the midsegments of the medial triangle created in the previous step and so on. The result is in figure. What is the total mass of Matěj's fractal?

Matěj likes infinities.

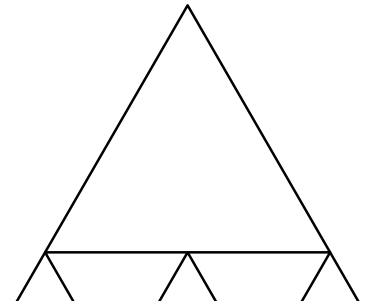


Problem X.2 ... just fractal

3 point

Matěj is still playing with the same fractal (with dimension $l = 10\,\mathrm{cm}$ *and linear density* $\lambda = 100\,\mathrm{g\cdot m^{-1}}$*). He would like to know its moment of inertia with respect to an axis that contains one of the medians of the largest triangle. Find this moment of inertia.*

Matěj really likes infinities.

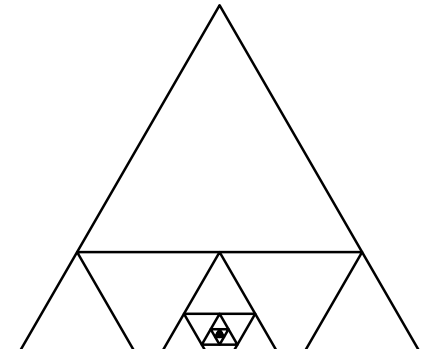


Problem X.3 ... quite fractal

3 point

Now, Matěj hanged his fractal (with dimension $l = 10\,\mathrm{cm}$ and linear density $\lambda = 100\,\mathrm{g\cdot m^{-1}}$) from one of the vertices of the largest triangle and let it perform small oscillations around this vertex in the plane that contains the fractal. What period of oscillations did Matěj measure?

Matěj really truly very much does like infinities.



Problem X.4 ... very fractal

3 point

Matěj still likes playing with his fractal (with dimension $l =$ $= 10\,\mathrm{cm}$). He found a multimeter and connected it to two vertices of the largest triangle. The resistance of one metre of wire is $000.000\,\Omega$. What resistance did Matěj measure?

Matěj really truly very much does like infinitely infinite infinities.

